

UNIT 1: Android OS & Development Environment

1. What is Android?
2. What is Android SDK?
3. What is an Android Virtual Device (AVD)?
4. Explain Android architecture.
5. What is Android Manifest file?

Coding

6. Write code to display “Hello Android” in an Activity.
 7. Write XML for a layout with TextView and Button.
 8. Write code to fetch a string from strings.xml.
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UNIT 2: Android Components & Resource Handling

9. What is Context?
10. What is an Intent?
11. Types of Intents with examples.
12. Difference between Activity and Service.
13. What is Broadcast Receiver?
14. What is Localization?

Coding

15. Write code to start a new Activity.
 16. Write an implicit Intent to open a website.
 17. Define and use a color resource in XML.
 18. Write a BroadcastReceiver skeleton.
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UNIT 3: UI Elements & Layouts

19. What is Material Design?
20. What are UI and UX?
21. Difference: LinearLayout vs RelativeLayout.
22. What is ConstraintLayout?
23. Types of menus in Android.
24. What are Dialog boxes?

- 25. XML for vertical LinearLayout.
 - 26. XML for ConstraintLayout with two buttons.
 - 27. Code to show AlertDialog.
 - 28. Handle button click event.
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UNIT 4: Views & Fragments

- 29. What is ListView?
- 30. What is Fragment?
- 31. RecyclerView vs ListView.
- 32. Fragment lifecycle.
- 33. ScrollView vs ListView.

Coding.

- 34. Add Fragment dynamically.
 - 35. XML for CardView.
 - 36. Implement Activity Life Cycle
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UNIT 5: Data Storage Techniques

- 37. What is SharedPreferences?
- 38. What is SQLite?
- 39. Explain Content Provider.
- 40. What is Content Resolver?

Coding

- 41. Save data using SharedPreferences.
- 42. Retrieve data from SharedPreferences.
- 43. SQLiteOpenHelper class skeleton.
- 44. Insert data into SQLite database.
- 45. Query data using Content Resolver.